

Task 1:

Project Name: UMass Marketplace

Group Name: Group 10

Group Members:

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Task 2: – Function Point Analysis (Frontend)

System Boundary:

The application boundary is the **UMass Marketplace frontend interface** developed for users to browse listings, sign in, filter items, view details, message sellers, and create listings. The frontend communicates with backend services and displays marketplace data to users. Since this writeup is focused on the frontend portion, the counted functions are based on the user-visible screens and interactions that I designed and implemented. The overall project and shared system entities are consistent with the team's UMass Marketplace design

Source of Function Counts:

These function point counts were derived from the frontend screens I worked on:

- Welcome page
- Google sign-in page
- Browse marketplace page
- Category filter dropdown
- Price filter dropdown
- Listing detail page
- Safety notice modal
- Messaging screen
- Create new listing screen

1. Internal Logical Files (ILF)

These are logical groups of data used by the frontend and maintained by the overall system database. The frontend interacts with them as core marketplace entities:

- **Users** – user profile and login identity
- **Listings** – item title, description, price, location, condition, seller, status

- **Categories** – category options shown in filters and listing creation
- **Messages / Conversations** – chat history between buyers and sellers
- **Photos** – listing images shown in browse, detail, and create listing screens

These are considered ILFs because they are user-recognizable data groups that are stored and maintained within the UMass Marketplace system, even though my work specifically focused on the frontend interface that displays and sends data related to them

2. External Interface Files (EIF)

- **Google Authentication Service** – the frontend includes a “Sign in with Google” flow, which depends on user data maintained by an external authentication provider rather than by the marketplace itself.

This is an EIF because the login process references data maintained outside the application boundary of the marketplace frontend/system.

3. External Inputs (EI)

These are user actions where data enters the system and causes a change to internal data or triggers a transaction.

- **Sign in with Google** – user initiates login/authentication
- **Search marketplace** – user enters search text
- **Select category filter** – user chooses a category
- **Select price filter** – user chooses a price range
- **Open message seller flow** – user initiates contact from listing page
- **Acknowledge safety notice** – user confirms understanding before proceeding
- **Send message** – user types and sends a chat message
- **Create listing** – enter title
- **Create listing** – enter price
- **Create listing** – choose category
- **Create listing** – enter location
- **Create listing** – choose condition
- **Create listing** – enter description
- **Upload photos**
- **Publish listing**

These are External Inputs because they involve data entered by the user through the interface and either update system state, create records, or submit information for processing.

4. External Outputs (EO)

These are outputs where the frontend presents processed information to the user, usually involving formatting, filtering, or combined display logic.

- **Marketplace browse results** – displays processed listings based on search/filter selections
- **Filtered listings by category** – shows listings after category-based filtering
- **Filtered listings by price** – shows listings after price-based filtering
- **Listing cards on browse page** – each card displays combined listing info such as image, price, title, and location
- **Listing detail page** – shows integrated listing information including seller, location, category, description, and images
- **Messaging conversation display** – shows chat history in a formatted conversation layout
- **Safety notice modal** – displays a formatted warning and interaction step before messaging
- **Listing publish confirmation/result** – result of submitting a new listing

These are External Outputs because the system sends processed or formatted information to the user, often combining multiple fields or applying filters.

5. External Inquiries (EQ)

These are retrieval actions where the user requests information and the system returns it without major computation.

- **View welcome page**
- **View sign-in page**
- **Open browse page**
- **Open category dropdown options**
- **Open price dropdown options**
- **View specific listing details**
- **View existing messages in a conversation**
- **Open create listing page**

These are External Inquiries because they primarily retrieve and display information without significant derived processing.

Function Point Calculation

Type	Count	Complexity	Weight	Total
ILF	5	Average	10	50
EIF	1	Average	7	7
EI	15	Average	4	60
EO	8	Average	5	40
EQ	8	Average	4	32

Total FP = 189

Unadjusted Function Points (UFP) = 189

Explanation of Complexity

All components were assumed to have **average complexity** because the frontend includes multiple interactive screens, moderate validation, filtered listing views, messaging functionality, listing creation, and integration with an external sign-in provider. The interface is more than a simple display-only UI because it supports both information retrieval and user-driven input workflows.

Summary

This frontend function point analysis reflects the functional size of the user-facing UMass Marketplace interface that I worked on. My screens support authentication, browsing, searching, filtering, viewing listing details, messaging sellers, and creating new listings. The analysis aligns with the overall marketplace design used by the team while focusing specifically on the frontend features and flows I implemented